

Conditional Exponential DLPP with Flat Initial Condition

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1-Dimensional Exponential Directed Last Passage Percolation

$$\mathcal{L}^{\text{ptp}}(m, 0) = \sum_{i=0}^m X_i, \quad X_i \sim \text{Exp}(1).$$

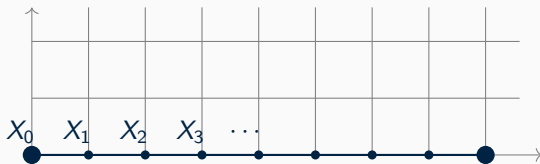


Figure 1: 1-D Case of Exponential DLPP

$$\sum_{i=0}^M X_i \xrightarrow{p} M, \quad \frac{\sum_{i=0}^M X_i - M}{\sqrt{M}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Exponential DLPP

$$E(\pi) := \sum_{(i,j) \in \pi} X_{i,j}, \quad \mathcal{L}^{\text{ptp}}(m, n) := \max_{\substack{\pi: (0,0) \rightarrow (m,n) \\ \pi \text{ up-right}}} E(\pi).$$

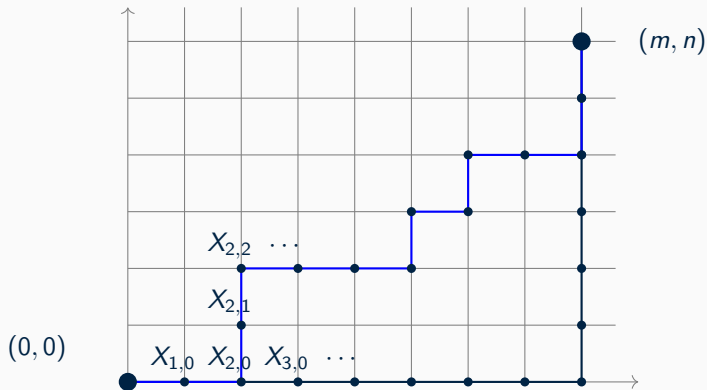


Figure 2: Exponential Directed Last Passage Percolation (DLPP)

Law of Large Numbers

Law of Large Numbers for Point-to-point DLPP [Rost 1981]:

$$\bar{\mathcal{L}}^{\text{ptp}}(x, y) := \lim_{N \rightarrow \infty} \frac{\mathcal{L}^{\text{ptp}}(xN, yN)}{N} \stackrel{p}{=} (\sqrt{x} + \sqrt{y})^2$$

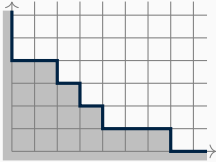


Figure 3: Corner Growth Model



Figure 4: Wildfire Front

Image source: Cox Media Group.

Results	References
Physics prediction	Huse–Henley (1985).
One-point fluctuations	Johansson (2000).
Functional fluctuations	Prähofer–Spohn (2002); Johansson (2003); Matetski–Quastel–Remenik (2021).

Upper Deviation Event

[Baik, Cordaro, Tripathi 2025] conditioned distribution upon the event that a given point (aN, bN) satisfies

$$\mathcal{L}^{ptp}(aN, bN) = \ell N > \bar{\mathcal{L}}^{ptp}(aN, bN).$$

What do we expect near the conditioned point?

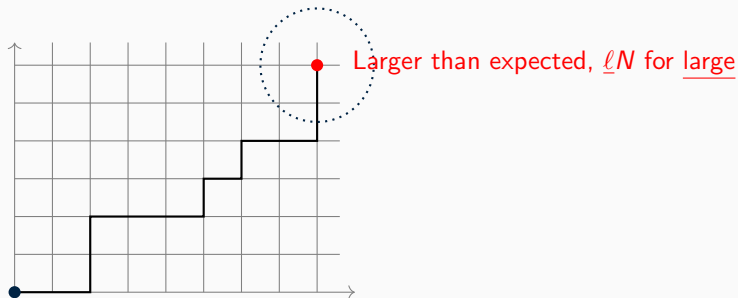


Figure 5: Exponential Directed Last Passage Percolation (DLPP)

Conditional Law of Large Number

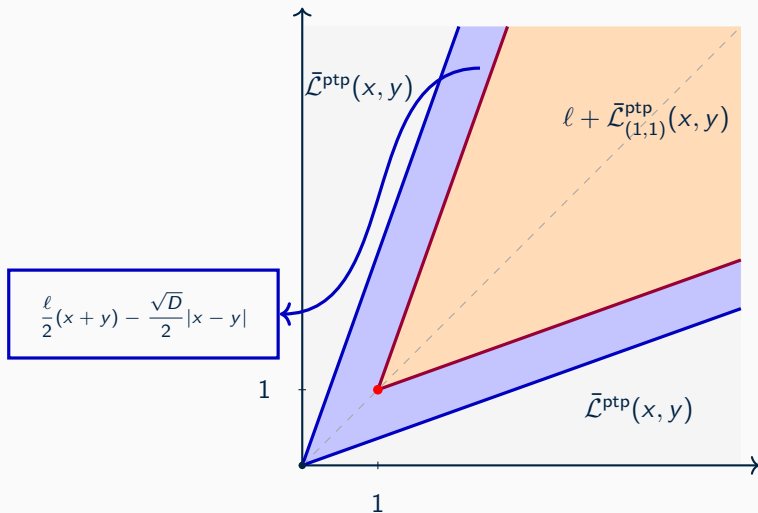


Figure 6: Three Law of Large Number Regimes in Point-to-Point Case

Our Setup: Flat Initial Condition

$$\mathcal{L}(m, n) := \max_{\pi: (-k, k) \rightarrow (m, n)} E(\pi).$$

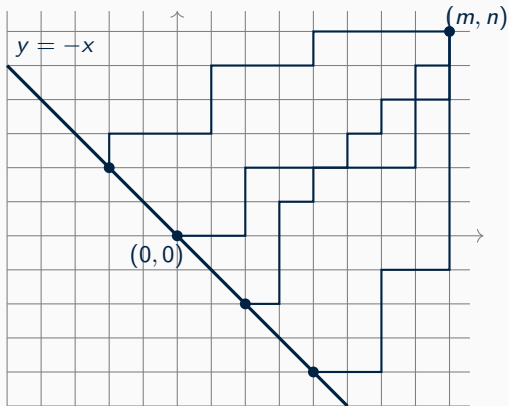


Figure 7: Flat Exponential DLPP

Part 1: Conditional Law of Large Numbers Heuristic

Part 2: Rigorous Computation of Fluctuations along Diagonal

Conditional Law of Large Numbers Heuristic

Conditioning on Upper Large Deviation Event

Law of Large Numbers for flat initial condition DLPP [Johansson 1999]:

$$\lim_{N \rightarrow \infty} \frac{\mathcal{L}(xN, yN)}{N} \stackrel{p}{=} \bar{\mathcal{L}}^{\text{flat}}(x, y)$$

Now we condition our distribution upon the event that a given point (aN, bN) satisfies

$$\mathcal{L}(aN, bN) = \ell N > \bar{\mathcal{L}}^{\text{flat}}(a, b)N.$$

That is,

$$\ell > \bar{\mathcal{L}}^{\text{flat}}(a, b).$$

We now follow a similar heuristic argument as in [Baik, Cordaro, Tripathi 2025].

Conditional Law of Large Numbers Heuristic

$$H(t) = t\ell + \bar{\mathcal{L}}_{(t,t)}^{\text{ptp}}(x, y)$$

$$\lim_{N \rightarrow \infty} \frac{\mathcal{L}(xN, yN)}{N} = \max\{\bar{\mathcal{L}}^{\text{flat}}(x, y), H(t) : 0 \leq t \leq \min(x, y, 1)\}^1$$

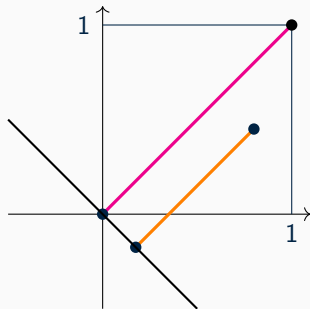


Figure 8: Unconditioned optimal starting points

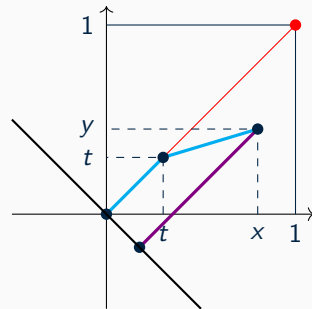


Figure 9: Conjectural maximizing path

¹Our expressions are under changes of coordinates.

Conditional Law of Large Numbers Heuristic (cont.)

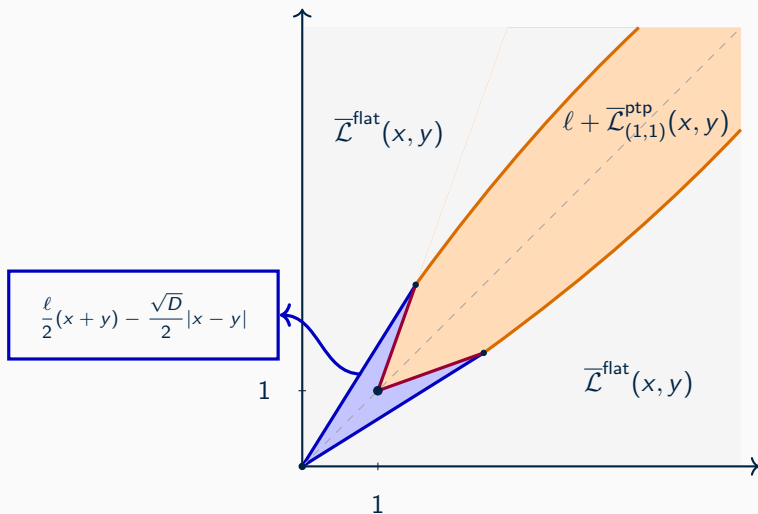


Figure 10: Three Law of Large Number Regimes in Flat Case

Compare with Point-to-Point Regimes

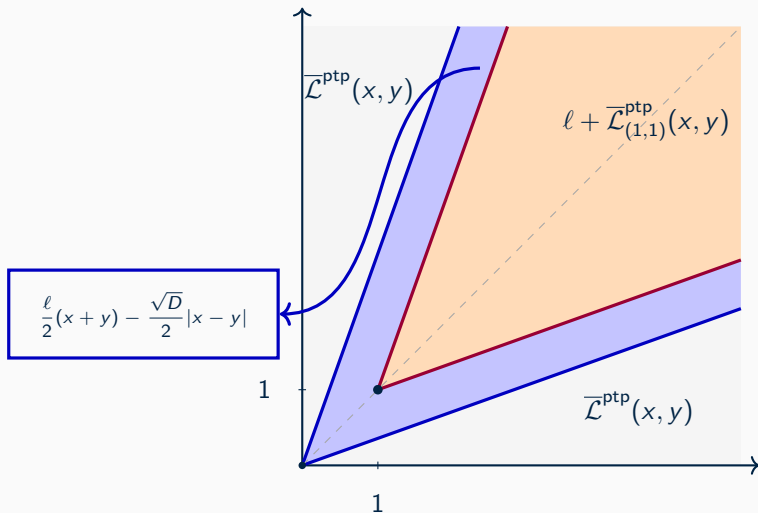
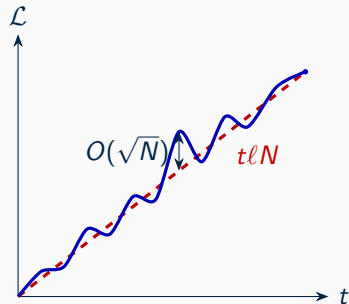


Figure 11: Three Law of Large Number Regimes in Step Case

Rigorous Computation of Fluctuations Along Diagonal

Limit shape and random fluctuation



Dashed: the deterministic $O(N)$ profile.

Blue: an $O(\sqrt{N})$ random deviation.

Comparison with the usual CLT

$$\frac{\sum_{i=1}^N X_i - N\mathbb{E}[X_i]}{\sigma\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1)$$

$$\frac{\mathcal{L}(\text{near}(tN, tN)) - t\ell N}{\sigma\sqrt{N}} \xrightarrow{d} X_N(t, s)$$

Intuitively, like Bayes' theorem:

$$\mathbb{P}(\mathcal{L}(M_1, N_1) > T_1 | \mathcal{L}(M_2, N_2) = T_2) = \frac{\frac{\partial}{\partial T_2} \mathbb{P}(\mathcal{L}(M_1, N_1) > T_1, \mathcal{L}(M_2, N_2) \leq T_2)}{\frac{\partial}{\partial T_2} \mathbb{P}(\mathcal{L}(M_2, N_2) \leq T_2)}.$$
²

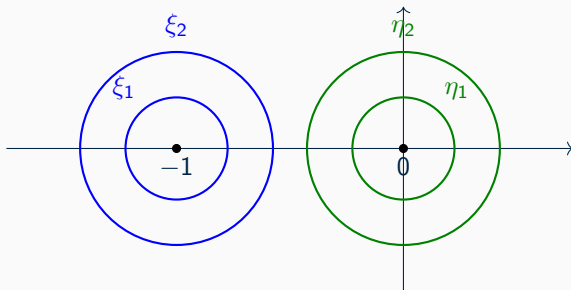
Focus on numerator, from [Liu 2021]:

$$\frac{\partial}{\partial T_2} \mathbb{P}(\mathcal{L}(M_1, N_1) > T_1, \mathcal{L}(M_2, N_2) \leq T_2) = \underbrace{Q_m^{(1)}(\mathbf{M}, \mathbf{N}, \mathbf{T})}_{\text{dominating term}} + \underbrace{(\text{series of integrals})}_{\text{bounded}}$$
³

²This is the 1-point distribution ($m=2$), we handle the multi-point distribution similarly in our research.

³This has been rewritten with notation as in [Baik, Cordaro, Tripathi 2025].

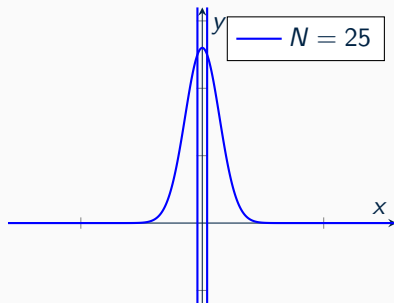
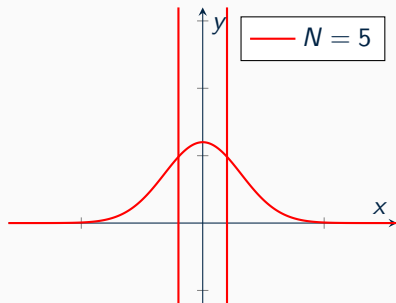
$$Q_m^{(1)}(\mathbf{M}, \mathbf{N}, \mathbf{T}) = - \left[\int_{C_{2,L}^{out}} \frac{d\xi_2}{2\pi i} \right] \left[\int_{C_{2,R}^{out}} \frac{d\eta_2}{2\pi i} \right] \left[\int_{C_{1,L}} \frac{d\xi_1}{2\pi i} \right] \left[\int_{C_{1,R}} \frac{d\eta_1}{2\pi i} \right] \underbrace{\Pi_1(\xi, \eta)}_{\text{Rational Function}} \underbrace{F_{\mathbf{M}, \mathbf{N}, \mathbf{T}}^{(1)}(\xi, \eta)}_{\text{Exponentials}} \underbrace{\delta(-\eta_1 - 1, \xi_1)}_{\text{Dirac } \delta\text{-kernel}}$$



Note: $\Pi_1(\xi, \eta)$ is written here with simpler notation

Laplace's Method

$$\lim_{N \rightarrow \infty} \left(\int_{-\delta_N}^{\delta_N} e^{N(G(x_{\text{crit}}) + \frac{1}{2}G''(x_{\text{crit}})x^2 + O(x^3))} dx + \int_{\mathbb{R} \setminus (-\delta_N, \delta_N)} e^{NG(x)} dx \right)$$



Contours for Method of Steepest Descent

We are applying steepest descent to $G(z) = -a \log(z + 1) + a \log(z) + \ell z$,

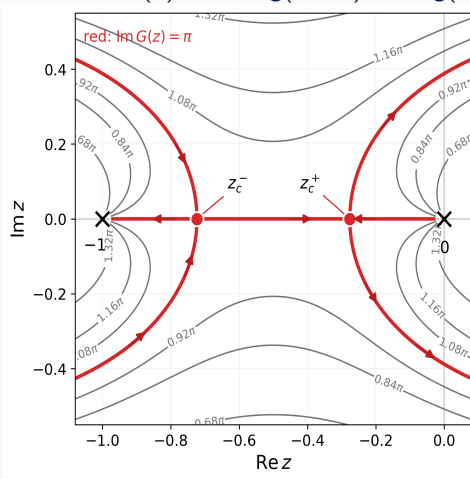
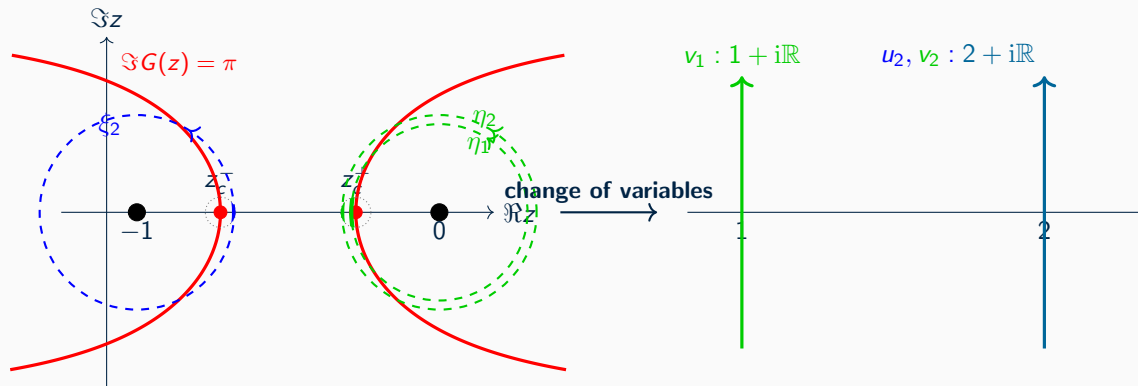


Figure 12: Level curves of $\text{Im } G(z)$.

Analysis near the Critical Points

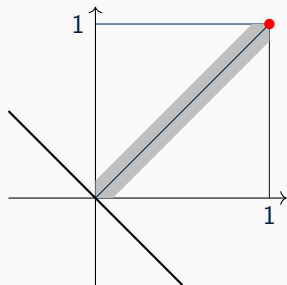


Fluctuations near Diagonal

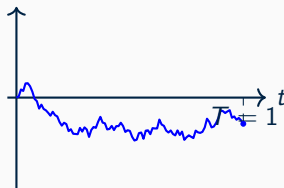
Theorem

Fix $\ell > \tilde{\mathcal{L}}(1, 1) = 4$. Then, as $N \rightarrow \infty$,

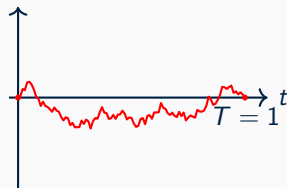
$$\text{Law} \left(\left\{ \frac{\mathcal{L}(tN + sCN^{1/2}, tN - sCN^{1/2}) - t\ell N}{\sigma N^{1/2}} \right\} \middle| \mathcal{L}(N, N) = \ell N \right) \xrightarrow{f.d.d.} \text{Law} (\mathbb{B}_1^{\text{br}}(t) - |\mathbb{B}_2^{\text{br}}(t) - (1-t)Z - s|)$$



Standard Brownian Motion



Brownian Bridge



Heuristic for General Fluctuations

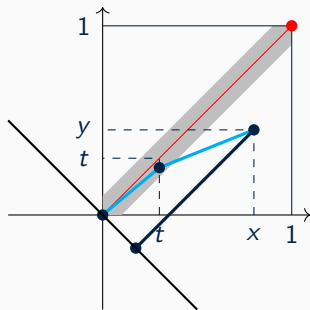


Figure 13: Conjectured general fluctuations.

$$\sigma N^{1/2} \begin{cases} \mathbb{B}_+(t_c) - \frac{1-t_c}{\sqrt{2}} Z, & y < x, \\ \mathbb{B}_-(t_c) + \frac{1-t_c}{\sqrt{2}} Z, & y > x. \end{cases}$$

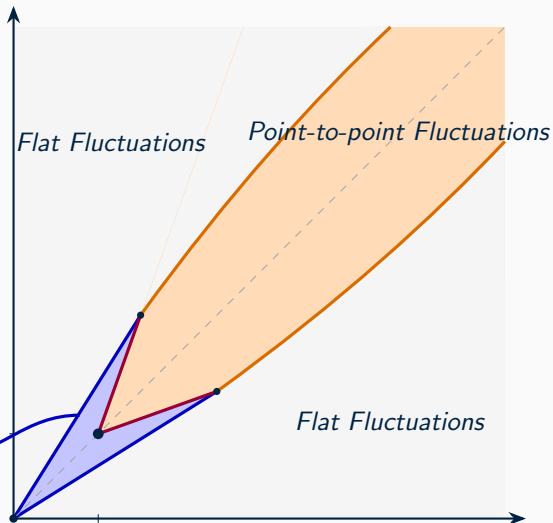
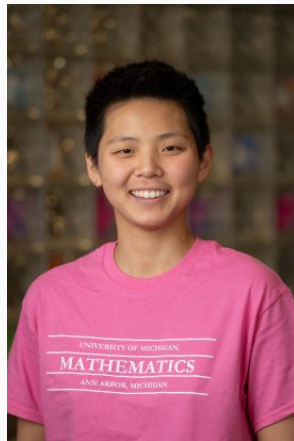


Figure 14: Three fluctuation regimes.

Acknowledgments



Prof. Baik



Tian Dong

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